

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 Assessment Tool 1 — Scenario-Based Requirement Analysis

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

for

AI-Based Smart Healthcare Appointment and Patient Flow Optimization Platform

Field	Details
Course Code / Title	CSA10 – Software Engineering
CO Assessed	CO2 – Perform detailed requirement analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills
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Signature of the Student: _____



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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for the AI-Based Smart Healthcare Appointment and Patient Flow Optimization Platform. The purpose of the system is to help hospitals and healthcare institutions manage appointment scheduling and patient flow intelligently by using artificial intelligence, predictive analytics, and cloud technologies, thereby reducing waiting times and improving the utilization of doctors, consultation rooms, and diagnostic resources.

This document is intended to serve as the foundation for design, development, testing, and validation of the proposed system. It has been prepared as part of the Scenario-Based Requirement Analysis assessment (CO2 – CSA10 Software Engineering) and follows the standard structure of an SRS as recommended by IEEE 830 guidelines, adapted to the given assessment format.

1.2 Document Conventions

Functional requirements are labelled FR-1, FR-2, ... and non-functional requirements are labelled NFR-1, NFR-2, ... for traceability. Priority levels (High / Medium / Low) indicate the relative importance of each requirement for the first release of the system.

1.3 Intended Audience

- Software architects and developers responsible for designing and building the platform.
- Hospital administrators and IT staff who will deploy, configure, and maintain the system.
- Quality assurance and testing teams validating system behaviour against requirements.
- Faculty evaluators assessing the requirement analysis exercise.

1.4 Business Domain and Stakeholders

The business domain is hospital outpatient management and clinical operations, specifically the scheduling of appointments and the flow of patients through departments, doctors, and diagnostic services. Healthcare institutions face rising patient volumes, long waiting times, and inefficient use of doctors and rooms because scheduling is largely manual and reactive rather than predictive.

Key Stakeholders

- Patients — seek convenient, timely appointments and shorter waiting times.
- Doctors / Clinicians — need optimized schedules and quick access to patient information.
- Receptionists / Front-desk staff — manage walk-ins, queues, and emergency prioritization.
- Hospital Administrators — responsible for resource planning, performance monitoring, and compliance.
- IT / System Administrators — maintain the platform, integrations, and data security.
- Regulatory bodies — enforce healthcare data protection and quality-of-care standards.

2. Problem Description

Hospitals often experience overcrowding due to inefficient appointment scheduling and unpredictable patient arrivals. Manual scheduling processes increase waiting times and reduce patient satisfaction. Healthcare administrators struggle to optimize doctor availability, consultation rooms, and diagnostic resources, and emergency cases are not always identified and prioritized quickly enough within the existing workflow.

An AI-driven appointment management platform can address these problems by predicting patient arrivals, optimizing scheduling decisions, prioritizing emergency cases, and providing real-time updates to patients and staff. Such a system can significantly improve hospital efficiency while ensuring a better overall patient experience.

Industry Context

Healthcare institutions experience increasing patient volumes that create long waiting times and inefficient resource utilization. Hospitals require intelligent appointment scheduling systems capable of balancing patient demand with doctor availability. Artificial intelligence, predictive analytics, and cloud technologies offer clear opportunities to optimize patient flow while improving healthcare quality and operational efficiency.



3. Scope of the System

The proposed system is a cloud-hosted, AI-assisted appointment and patient-flow management platform accessible via web and mobile interfaces. It covers appointment booking and management, AI-based prediction of patient arrival patterns, optimization of doctor and resource allocation, telemedicine appointment support, real-time notifications, and hospital performance analytics.

3.1 In Scope

- Patient registration, authentication, and profile management.
- Intelligent appointment scheduling with AI-based arrival prediction.
- Doctor availability management and resource (room/equipment) allocation optimization.
- Emergency case detection and queue prioritization.
- Telemedicine appointment scheduling and virtual-consultation support.
- Real-time notifications to patients and staff (SMS/email/push).
- Hospital performance analytics and administrative dashboards.
- Integration with existing Hospital Information Systems (HIS) / Electronic Health Records (EHR).

3.2 Out of Scope

- Clinical diagnosis, treatment recommendation, or prescription generation (the system supports operations, not clinical decision-making).
- Billing and insurance claim processing beyond basic payment confirmation for telemedicine sessions.
- Inpatient (ward/bed) management — the system is limited to outpatient appointment and flow management in the first release.



4. Functional Requirements

Functional requirements are grouped by module for clarity. Each requirement is uniquely identified for traceability against objectives and test cases.

4.1 User Management

ID	Requirement	Priority
FR-1	The system shall allow patients to register and create a profile with personal and basic medical details.	High
FR-2	The system shall allow patients, doctors, receptionists, and administrators to securely log in using role-based credentials (e.g., password and OTP).	High
FR-3	The system shall allow administrators to create, modify, deactivate, and assign roles/permissions to user accounts.	Medium

4.2 Appointment Scheduling

ID	Requirement	Priority
FR-4	The system shall allow patients to book appointments by selecting department, doctor, preferred date, and time slot.	High
FR-5	The system shall use an AI prediction model to forecast patient arrival patterns based on historical appointment and walk-in data.	High
FR-6	The system shall automatically recommend optimal appointment slots based on doctor availability and predicted patient load.	High
FR-7	The system shall allow patients and staff to reschedule or cancel existing appointments.	High
FR-8	The system shall detect and flag emergency cases and reprioritize the scheduling queue accordingly.	High

4.3 Doctor and Resource Allocation

ID	Requirement	Priority
FR-9	The system shall allow administrators to manage doctor schedules, consultation rooms, and diagnostic resource availability.	High
FR-10	The system shall optimize allocation of doctors and rooms based on real-time and forecasted patient demand.	Medium
FR-11	The system shall allow doctors to update their availability, leave status, and consultation hours in real time.	Medium

4.4 Telemedicine

ID	Requirement	Priority
FR-12	The system shall support scheduling of virtual (telemedicine)	Medium



ID	Requirement	Priority
	consultations alongside in-person appointments.	
FR-13	The system shall integrate with a third-party video-conferencing service to conduct remote consultations.	Medium
FR-14	The system shall allow doctors to access relevant patient history during a telemedicine session.	Medium

4.5 Notifications

ID	Requirement	Priority
FR-15	The system shall send real-time notifications (SMS/email/push) for appointment confirmation, reminders, and delays.	High
FR-16	The system shall notify patients of their live queue position and estimated waiting time.	Medium
FR-17	The system shall alert relevant staff immediately when an emergency case is registered.	High

4.6 Analytics and Reporting

ID	Requirement	Priority
FR-18	The system shall generate hospital performance analytics covering patient flow, doctor utilization, and average waiting time.	Medium
FR-19	The system shall provide administrators with a real-time dashboard of patient flow and resource utilization.	Medium
FR-20	The system shall generate periodic (daily / weekly / monthly) operational reports.	Low

4.7 Feedback

ID	Requirement	Priority
FR-21	The system shall allow patients to submit feedback and a satisfaction rating after a consultation.	Low
FR-22	The system shall maintain an audit log of scheduling changes, cancellations, and priority overrides for accountability.	Medium



5. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	The system shall compute an AI-based scheduling recommendation within 3 seconds under normal load, and shall support at least 5,000 concurrent users across a multi-hospital deployment.
NFR-2	Security	The system shall enforce role-based access control, encrypt patient data at rest and in transit (AES-256 / TLS 1.2+), and maintain compliance with applicable healthcare data protection regulations.
NFR-3	Reliability	The system shall achieve at least 99.5% uptime, excluding scheduled maintenance windows, and shall recover from failure with no loss of confirmed appointment data.
NFR-4	Usability	The system shall provide an intuitive, accessible interface (WCAG 2.1 AA) usable by patients, doctors, and staff with minimal training, and shall support at least two languages (English and a regional language).
NFR-5	Scalability	The system architecture shall support horizontal scaling to accommodate additional hospital branches, departments, and user growth without redesign.
NFR-6	Maintainability	The system shall be built using a modular, well-documented architecture (e.g., microservices with versioned APIs) to simplify updates and defect resolution.
NFR-7	Availability	The system shall be available 24/7, with planned maintenance restricted to low-usage hours and communicated in advance to users.
NFR-8	Interoperability	The system shall exchange patient and appointment data with existing Hospital Information Systems / EHR platforms using standard protocols (e.g., HL7 or FHIR).
NFR-9	Data Privacy & Compliance	The system shall handle patient data in compliance with applicable regulations (e.g., India's DPDP Act, HIPAA/GDPR where applicable) including consent management and data retention limits.



6. Actors and Use Cases

6.1 Actor Descriptions

Actor	Type	Description
Patient	Primary	Registers, books/reschedules/cancels appointments, receives notifications, views queue status, and joins telemedicine consultations.
Receptionist / Front-Desk Staff	Primary	Registers walk-in patients, manages the appointment queue, and flags emergency cases.
Doctor	Primary	Views schedule, updates availability, accesses patient records, and conducts in-person or telemedicine consultations.
Hospital Administrator	Primary	Manages doctor and resource allocation, monitors performance analytics, and administers user accounts and system configuration.
SMS / Email Gateway	Secondary (External System)	Delivers real-time notifications and reminders to patients and staff on behalf of the system.
Telemedicine Platform	Secondary (External System)	Hosts video-conferencing sessions for remote consultations, invoked by the system.
Hospital Information System (EHR)	Secondary (External System)	Provides and receives patient medical records and history for interoperability.

6.2 Use Case Diagram

The diagram below shows the primary actors (Patient, Receptionist, Doctor, Hospital Administrator), the secondary/external systems (SMS/Email Gateway, Telemedicine Platform, Hospital Information System), and the major use cases of the platform.



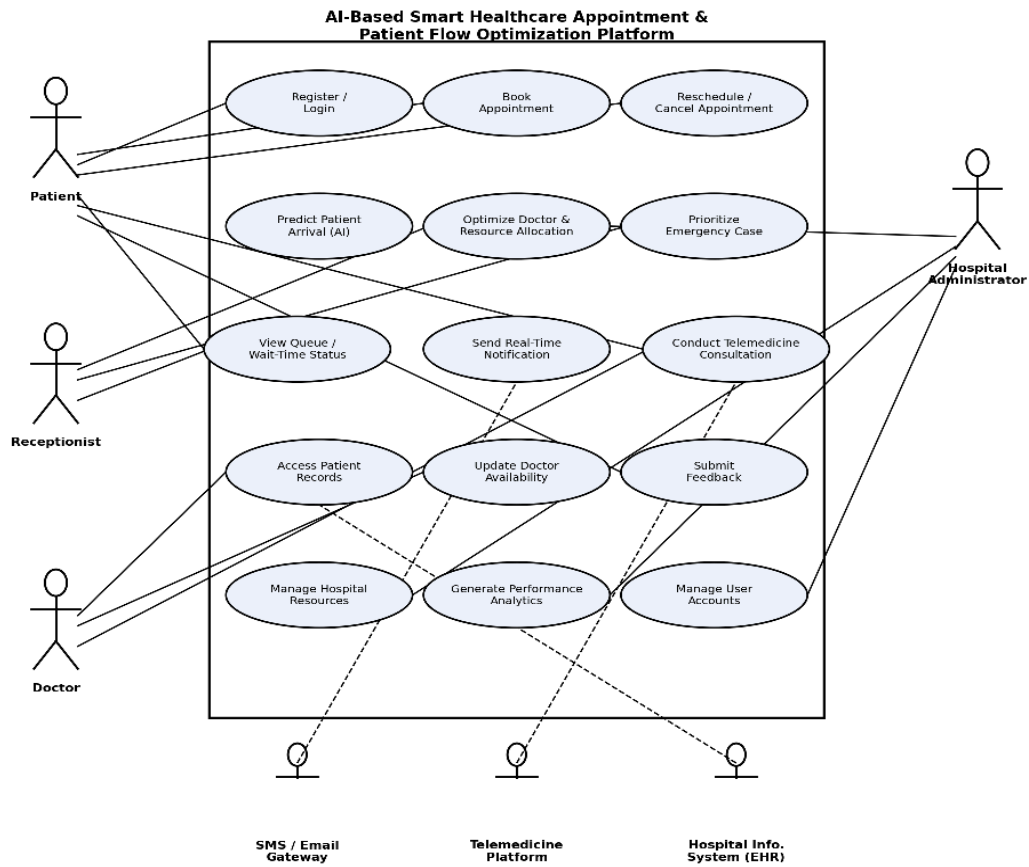


Figure 1: Use Case Diagram — AI-Based Smart Healthcare Appointment and Patient Flow Optimization Platform.

6.3 Use Case Descriptions

UC-1: Book Appointment

Field	Description
Actor(s)	Patient (primary), AI Scheduling Engine, SMS/Email Gateway (secondary)
Description	A patient searches for a department/doctor and books an appointment for a preferred date and time.
Preconditions	The patient is registered and logged in.
Main Flow	1) Patient selects department and doctor. 2) System displays AI-recommended available slots. 3) Patient selects a slot and confirms. 4) System reserves the slot and sends a confirmation notification.
Alternate Flow	If the preferred slot is unavailable, the system suggests the nearest optimal alternative slot based on predicted demand.
Postconditions	An appointment record is created and the patient is notified.

UC-2: Predict Arrival & Optimize Schedule (AI)

Field	Description
Actor(s)	AI Scheduling Engine (system function), Hospital Administrator
Description	The system analyzes historical and real-time data to predict patient arrival patterns and recommends optimal doctor/room allocation.
Preconditions	Sufficient historical appointment/attendance data is available.
Main Flow	1) System collects historical and live scheduling data. 2) AI model predicts expected arrivals per time slot. 3) System recalculates optimal doctor/room allocation. 4) Administrator reviews and approves/adjusts recommendations.
Alternate Flow	If predicted demand exceeds available capacity, the system flags the time slot for administrator review and suggests reallocating resources.
Postconditions	Updated schedule and resource allocation are published to relevant staff.

UC-3: Conduct Telemedicine Consultation

Field	Description
Actor(s)	Patient, Doctor (primary), Telemedicine Platform (secondary)
Description	A doctor and patient conduct a scheduled remote consultation through the integrated video platform.
Preconditions	A telemedicine appointment has been booked and confirmed.
Main Flow	1) System sends a join link to patient and doctor at the scheduled time. 2) Both parties join the video session. 3) Doctor accesses patient history during the call. 4) Doctor marks the consultation complete and adds notes.
Alternate Flow	If the patient does not join within a defined grace period, the system marks the appointment as a no-show and notifies the doctor.



Field	Description
Postconditions	Consultation record is updated; patient may submit feedback.

UC-4: Prioritize Emergency Case

Field	Description
Actor(s)	Receptionist (primary), Doctor
Description	A walk-in emergency patient is registered and inserted into the queue ahead of routine appointments.
Preconditions	Receptionist has access to the queue management module.
Main Flow	1) Receptionist registers the walk-in patient and marks the case as emergency. 2) System recalculates queue order, prioritizing the emergency case. 3) System notifies the assigned doctor and affected patients of any delay. 4) Doctor attends to the emergency case.
Alternate Flow	If no doctor is immediately available, the system identifies the nearest available doctor/room and reallocates accordingly.
Postconditions	Queue order is updated and audit log entry is recorded (FR-22).

UC-5: Generate Hospital Performance Analytics

Field	Description
Actor(s)	Hospital Administrator (primary)
Description	The administrator views or generates analytics on patient flow, doctor utilization, and waiting times.
Preconditions	Administrator is logged in with appropriate permissions.
Main Flow	1) Administrator opens the analytics dashboard. 2) System aggregates appointment, queue, and utilization data. 3) System displays key metrics and trends. 4) Administrator exports or schedules a periodic report.
Alternate Flow	If data for a selected period is incomplete, the system indicates partial results and the underlying cause.
Postconditions	Report is generated/exported for review or compliance records.



7. Data Requirements

Data Entity	Key Attributes
Patient	Patient ID, name, date of birth, contact details, medical history summary, appointment history, consent records.
Doctor	Doctor ID, name, specialization, qualifications, schedule, availability status, consultation room assignment.
Appointment	Appointment ID, patient ID, doctor ID, department, date/time, mode (in-person/telemedicine), status, priority flag.
Resource	Resource ID, type (room/equipment), location, availability status, current assignment.
Queue Entry	Queue ID, patient ID, arrival time, priority level, estimated wait time, status.
Notification Log	Notification ID, recipient, channel (SMS/email/push), message type, timestamp, delivery status.
Analytics Snapshot	Metric type, department, time period, aggregated value, generated timestamp.
Audit Log	Log ID, user, action type, affected record, timestamp, reason/justification.

Data Retention & Backup: Patient and appointment records shall be retained per applicable healthcare regulations and hospital policy, with automated daily backups and a defined disaster-recovery point objective (RPO) of no more than 24 hours.



8. External Interface Requirements

8.1 User Interfaces

- Patient mobile app / web portal for registration, booking, notifications, and queue status.
- Doctor dashboard for schedule, patient history access, and telemedicine sessions.
- Receptionist console for walk-in registration and queue/priority management.
- Administrator dashboard for resource management, analytics, and system configuration.

8.2 Hardware Interfaces

- Optional self-service check-in kiosks at hospital entrances.
- Digital display boards showing live queue status in waiting areas.

8.3 Software Interfaces

- Hospital Information System (HIS) / EHR integration via HL7/FHIR for patient records.
- SMS / Email gateway API for notifications.
- Third-party video-conferencing API for telemedicine sessions.
- AI/ML prediction service for arrival forecasting and schedule optimization.
- Payment gateway API for telemedicine consultation fees (where applicable).

8.4 Communication Interfaces

- Secure HTTPS/REST APIs for all client-server communication.
- HL7/FHIR messaging standards for clinical data exchange with external systems.



9. System Features

Feature	Related Objective
Intelligent Appointment Scheduling Engine	Develop an intelligent appointment scheduling system.
AI-Based Patient Arrival Prediction	Predict patient arrival patterns using AI.
Doctor & Resource Allocation Optimizer	Optimize doctor allocation and consultation schedules.
Emergency Prioritization Module	Reduce patient waiting time / improve emergency prioritization.
Telemedicine Integration	Integrate telemedicine appointment support.
Real-Time Notification System	Provide real-time notifications to patients.
Hospital Performance Analytics Dashboard	Generate hospital performance analytics.
Resource Utilization Reporting	Improve utilization of healthcare resources.



10. Assumptions and Constraints

10.1 Assumptions

- Hospitals have stable internet connectivity to support cloud-based operation.
- Patients have access to a smartphone or internet-enabled device for booking and receiving notifications.
- Sufficient historical patient-flow data is available to train and validate the AI prediction model.
- Doctors and administrative staff will receive adequate training before go-live.
- Existing Hospital Information Systems expose interfaces (or can be adapted) for integration.

10.2 Constraints

Type	Constraint
Technical	The system must integrate with legacy hospital systems that may use outdated data formats or lack modern APIs.
Operational	24/7 operation requires redundant infrastructure and on-call support staff, increasing operational overhead.
Economic	Budget limitations may restrict the initial sophistication of the AI model and the scale of cloud infrastructure.
Legal / Regulatory	The system must comply with healthcare data protection regulations (e.g., India's DPDP Act, and HIPAA/GDPR where relevant to the deployment region).
Time	Implementation must be completed within the project/academic timeline, limiting the scope of the first release.



11. Requirement Validation

11.1 Traceability: Objectives vs. Functional Requirements

Objective	Addressed By
Develop an intelligent appointment scheduling system.	FR-4, FR-6
Predict patient arrival patterns using AI.	FR-5
Optimize doctor allocation and consultation schedules.	FR-9, FR-10, FR-11
Reduce patient waiting time.	FR-6, FR-8, FR-16
Integrate telemedicine appointment support.	FR-12, FR-13, FR-14
Provide real-time notifications to patients.	FR-15, FR-16, FR-17
Generate hospital performance analytics.	FR-18, FR-19, FR-20
Improve utilization of healthcare resources.	FR-9, FR-10, FR-18

All eight stated objectives are traceable to at least one functional requirement, confirming completeness of coverage at the objective level.

11.2 Completeness and Consistency Check

- Every stakeholder identified in Section 1.4 has at least one corresponding actor and use case (Section 6), confirming stakeholder coverage.
- FR-7 (reschedule/cancel) and FR-8 (emergency prioritization) were reviewed together to ensure they do not conflict: emergency reprioritization (FR-8) takes precedence over standard reschedule requests (FR-7), and this precedence rule has been made explicit to avoid ambiguity.
- No functional requirement was found to contradict a non-functional requirement; for example, the 3-second AI response target (NFR-1) is consistent with the real-time notification requirement (FR-15/FR-16).

11.3 Ambiguity and Redundancy Review

- The term “real-time” was ambiguous in the original problem statement; it has been quantified in NFR-1 (3 seconds for AI recommendations) and NFR-7 (24/7 availability) to remove subjectivity.
- FR-15 and FR-16 initially appeared redundant (both are notifications); they have been distinguished by content — FR-15 covers appointment lifecycle events while FR-16 covers live queue position — so both are retained as distinct requirements.
- No duplicate or conflicting requirements remained after this review.

11.4 Missing Requirements Identified and Added

- FR-22 (audit logging) was added after review, since emergency prioritization and schedule

overrides require accountability that was not captured in the original objective list.

- NFR-9 (data privacy & compliance) was made explicit, since handling patient medical data inherently requires regulatory compliance even though it was not stated as a separate objective.
- NFR-4 usability was extended to include accessibility (WCAG 2.1 AA) and multilingual support, given the diverse patient population expected to use the platform.

11.5 Suggested Improvements

- Define explicit Service Level Agreements (SLAs) for notification delivery time and AI recommendation latency.
- Add a formal data-backup and disaster-recovery schedule to the non-functional requirements (partially captured in Section 7).
- Conduct a pilot rollout in a single department before hospital-wide deployment, to validate AI prediction accuracy against real operational data.
- Establish a periodic requirement review cycle as new hospital departments or integrations are added.



12. Conclusion

This Software Requirement Specification captures the functional and non-functional requirements, actors and use cases, data and interface requirements, and assumptions/constraints for the AI-Based Smart Healthcare Appointment and Patient Flow Optimization Platform. The requirement set has been validated for completeness, consistency, and traceability against the stated objectives and expected outcomes, and gaps identified during validation (audit logging, data privacy, accessibility) have been incorporated. This document provides a sound basis for subsequent system design and development.

